



Chapter 1 - Introduction

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1. What is Data Mining?

- We are actually living in the data age.
 - Businesses worldwide generate gigantic data sets, including sales transactions, stock trading records, product descriptions, sales promotions, company profiles and performance, and customer feedback.
 - Scientific and engineering practices generate high orders of petabytes of data in a continuous manner, from remote sensing, to process measuring, scientific experiments, system performance, engineering observations, and environment surveillance.
 - Biomedical research and the health industry generate tremendous amounts of data from gene sequence machines, biomedical experiment and research reports, medical records, patient monitoring, and medical imaging.
 - **The list of sources that generate huge amounts of data is endless.**



1. What is Data Mining?

- Essentially, *data mining is the process of discovering interesting patterns, models, and other kinds of knowledge in large data sets.*
 - Knowledge mining from data
 - KDD (i.e., Knowledge Discovery from Data), pattern discovery, knowledge extraction, data archaeology, data analytics, and information harvesting.
- Data mining turns a large collection of data into knowledge.

2. Data Mining: An Essential Step in Knowledge Discovery

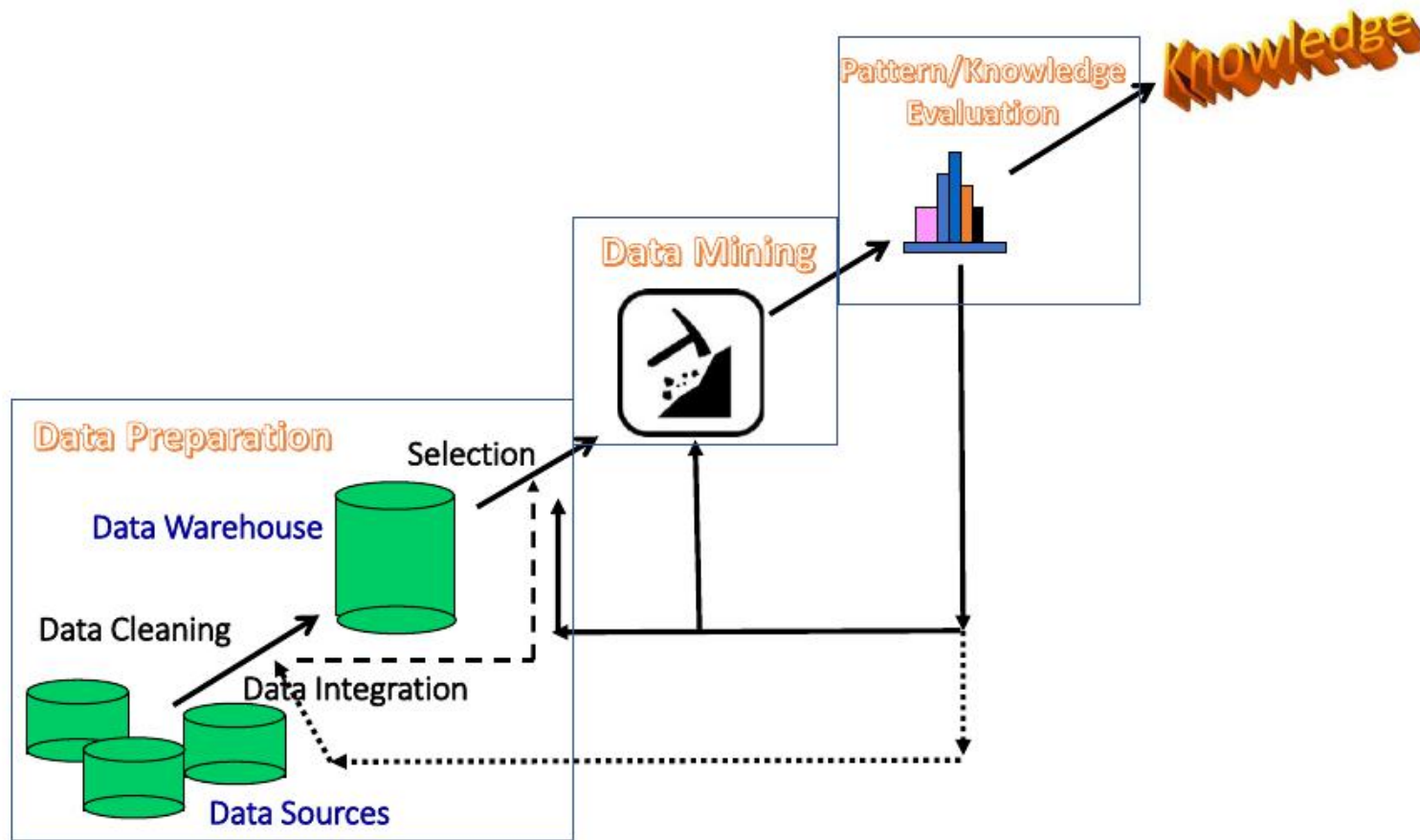


FIGURE 1.1

Data mining: An essential step in the process of knowledge discovery.

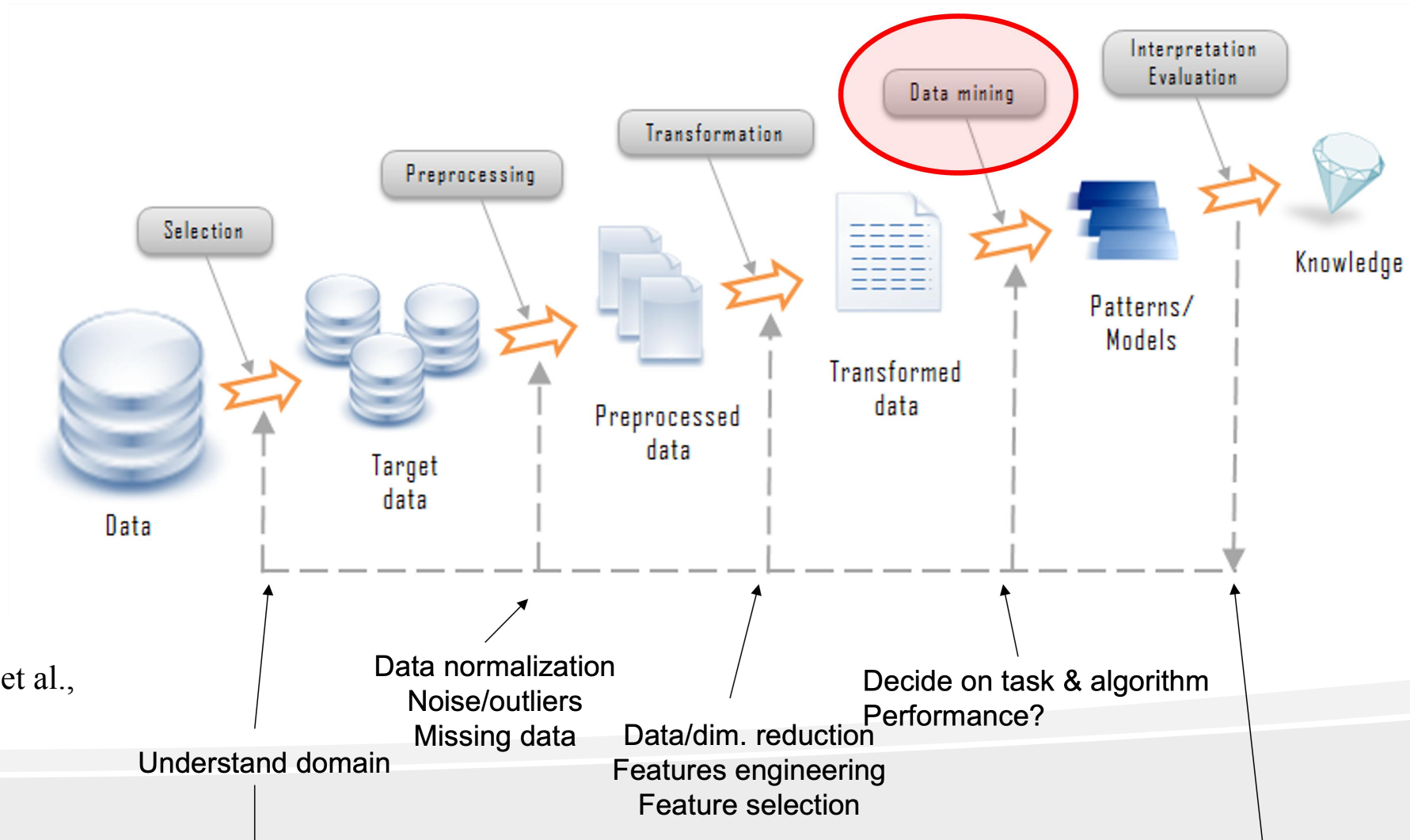


2. Data Mining: An Essential Step in Knowledge Discovery

- 1. Data Preparation
 - a. Data cleaning (to remove noise and inconsistent data)
 - b. Data integration (where multiple data source may be combined)
 - c. Data transformation (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
- 2. **Data Mining** (an essential process where intelligent methods are applied to extract patterns or construct models)
- 3. Pattern/model evaluation (to identify the truly interesting patterns or models representing knowledge based on *interestingness measures*)
- 4. Knowledge presentation (where visualization and knowledge representation techniques are used to present mined knowledge to users)

However, the term *data mining* is often used to refer to the entire knowledge discovery process (perhaps because the term is shorter than *knowledge discovery from data*).

2. Data Mining: An Essential Step in Knowledge Discovery



Usama M. Fayyad et al.,
1996



3. Diversity of Data Types for Data Mining

- Data mining can be applied to any kind of data as long as the data are meaningful for a target application.
 - However, different kinds of data may need rather different data mining methodologies.
- Structured vs. unstructured data
 - Based on whether data have clear structures, we can categorize data as structured vs. unstructured data.
- Data associated with different applications
 - Different applications may generate or need to handle very different data sets and require rather different data analysis methods.
 - Thus when categorizing data sets for data mining, we should take specific applications into consideration.
- Stored vs. streaming data
 - Usually, data mining handles finite, stored data sets, such as those stored in various kinds of large data repositories. However, in some applications such as video surveillance or remote sensing, data may stream in dynamically and constantly, as infinite data streams.



4. Mining Various Kinds of Knowledge

- In general, data mining tasks can be put into two categories: **descriptive data mining** and **predictive data mining**.
- Descriptive mining characterizes properties of the interested set of data, whereas predictive mining performs induction on the data set in order to make predictions.



4. Mining Various Kinds of Knowledge

Multidimensional Data Summarization

- It is often tedious for a user to go over the details of a large set of data. Thus it is desirable to automatically summarize an interested set of data and compare it with the contrasting sets at some high levels.
- Such summaritive description of an *interested set of data* is called **data summarization**.
- Data summarization can often be conducted in a multidimensional space.
 - product category, producer, location, or time, ...
 - The output of such multidimensional summarization can be presented in various forms, such as pie charts, bar charts, curves, multidimensional data cubes, and multidimensional tables, including crosstabs.



4. Mining Various Kinds of Knowledge

Mining Frequent Patterns, Associations, and Correlations

- **Frequent patterns** are patterns that occur frequently in data.
 - Mining frequent patterns leads to the discovery of interesting associations and correlations within data.
 - A *frequent itemset* typically refers to a set of items that often appear together in a transactional data set—for example, milk and bread, which are frequently bought together in grocery stores by many customers.
 - *Frequent itemset mining* is a fundamental form of frequent pattern mining.



4. Mining Various Kinds of Knowledge

Mining Frequent Patterns, Associations, and Correlations

- **Association analysis**

- Suppose that, a webstore manager wants to know which items are frequently purchased together (i.e., in the same transaction). An example of such a rule, mined from the transactional database, is
 - $buys(X, \text{"computer"}) \Rightarrow buys(X, \text{"webcam"})$ [*support* = 1%, *confidence* = 50%],
 - X is a variable representing a customer.
 - A **confidence**, or certainty, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy webcam as well.
 - A 1% **support** means that 1% of all the transactions under analysis show that computer and webcam are purchased together.
 - Association rules that contain a single predicate (*buys*) are referred to as **single-dimensional association rules**.
 - Dropping the predicate notation, the rule can be written simply as “*computer* $\Rightarrow$ *webcam* [1%, 50%].”



4. Mining Various Kinds of Knowledge

Mining Frequent Patterns, Associations, and Correlations

- Suppose, mining the same database generates another association rule:
 - $age(X, "20..29") \wedge income(X, "40K..49K") \Rightarrow buys(X, "laptop")$ [*support* = 0.5%, *confidence* = 60%].
 - Each attribute is referred to as a dimension, the above rule can be referred to as a **multidimensional association rule**.
- Typically, association rules are discarded as uninteresting if they do not satisfy both a **minimum support threshold** and a **minimum confidence threshold**.
- Additional analysis can be performed to uncover interesting statistical **correlations** between associated attribute–value pairs.



4. Mining Various Kinds of Knowledge

Classification and Regression for Predictive Analysis

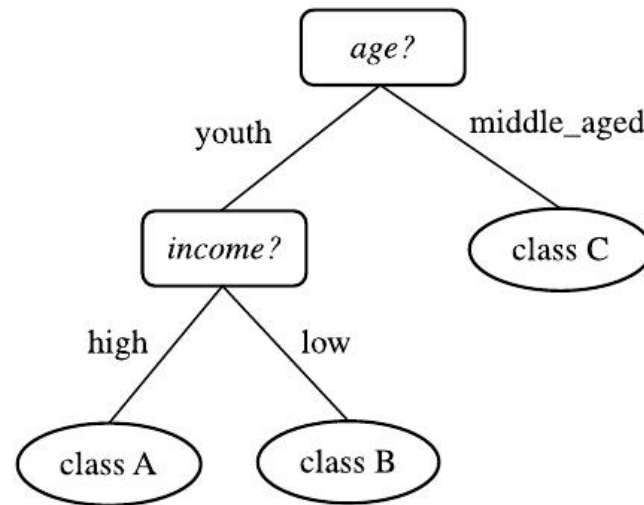
- **Classification** is the process of finding a **model** (or function) that describes and distinguishes data classes or concepts. The model is derived based on the analysis of a set of **training data** (i.e., data objects for which the class labels are known).
 - The model is used to predict the class labels of objects for which the class labels are unknown.
- Depending on the classification methods, a derived model can be in various forms, such as a set of *classification rules* (i.e., *IF-THEN rules*), a *decision tree*, a *mathematical formula*, or a learned *neural network* (Fig. 1.2).
 - There are many other methods for constructing classification models, such as naïve Bayesian classification, support vector machines, and *k*-nearest-neighbor classification.

4. Mining Various Kinds of Knowledge

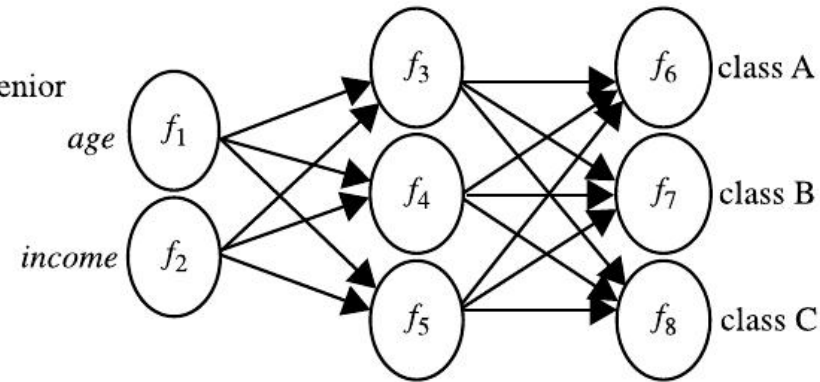
Classification and Regression for Predictive Analysis

$age(X, \text{"youth"}) \text{ AND } income(X, \text{"high"}) \longrightarrow class(X, \text{"A"})$
 $age(X, \text{"youth"}) \text{ AND } income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$
 $age(X, \text{"middle_aged"}) \longrightarrow class(X, \text{"C"})$
 $age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$

(a)



(b)



(c)

FIGURE 1.2

A classification model can be represented in various forms: (a) IF-THEN rules, (b) a decision tree, or (c) a neural network.



4. Mining Various Kinds of Knowledge

Classification and Regression for Predictive Analysis

- **Regression**

- Whereas classification predicts categorical (discrete, unordered) labels, **regression** models continuous-valued functions.
- That is, regression is used to predict missing or unavailable numerical data values rather than (discrete) class labels.
- The term *prediction* refers to both numeric prediction and class label prediction.
- **Regression analysis** is a statistical methodology that is most often used for numeric prediction, although other methods exist as well. Regression also encompasses the identification of distribution *trends* based on the available data.



4. Mining Various Kinds of Knowledge

Cluster Analysis

- **Cluster analysis** (also called **clustering**) groups data objects without consulting class labels.
 - In many cases, class-labeled data may simply not exist at the beginning. Clustering can be used to generate class labels for a group of data.
 - The objects are clustered or grouped based on the principle of maximizing the intraclass similarity and minimizing the interclass similarity. That is, clusters of objects are formed so that objects within a cluster have high similarity in comparison to one another, but are rather dissimilar to objects in other clusters.
 - Each cluster so formed can be viewed as a class of objects, from which rules can be derived.
- Clustering can also facilitate taxonomy formation, that is, the organization of observations into a hierarchy of classes that group similar events together.



4. Mining Various Kinds of Knowledge

Deep Learning

- For many data mining tasks, such as classification and clustering, a key step often lies in finding “good features,” which is a vector representation of each input data tuple.
 - Traditionally, this step (called feature engineering) often heavily relies on domain knowledge.
- Deep learning techniques provide an automatic way for feature engineering, which is capable of generating semantically meaningful features from the initial input features. The generated features often significantly improve the mining performance (e.g., classification accuracy).
- Deep learning is based on *neural networks*.



4. Mining Various Kinds of Knowledge

Outlier Analysis

- A data set may contain objects that do not comply with the general behavior or model of the data. These data objects are **outliers**. Many data mining methods discard outliers as noise or exceptions.
- However, in some applications (e.g., fraud detection) the rare events can be more interesting than the more regularly occurring ones. The analysis of outlier data is referred to as **outlier analysis** or **anomaly mining**.
- Outliers may be detected using statistical tests that assume a distribution or probability model for the data, or using distance measures where objects that are remote from any other cluster are considered outliers.
- Rather than using statistical or distance measures, density-based methods may identify outliers in a local region, although they look normal from a global statistical distribution view.